

Department of Electronic & Telecommunication Engineering, University of Moratuwa, Sri Lanka.

Assignment 3

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Submitted in partial fulfillment of the requirements for the module

EN3330 Introduction to Engineering Optimization

18th October 2026

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1 Introduction

This assignment investigates constrained optimization using the Lagrangian and duality approaches. The Lagrangian, dual function, and primal function are derived and plotted for the given optimization problem. The saddle-point condition, complementary slackness, and weak duality are also examined to understand the relationship between primal and dual solutions.

The Python codes used to generate the plots in this report are available in the following GitHub repository: [https://github.com/dahamif/Engineering-Optimization].

2 Lagrangian, Dual and Primal Functions

2.1 Plotting the Lagrangian

The Lagrangian function

$$L(\mu, x) = x^2 + \mu(1 - x)$$

was plotted against x for $\mu = 0, 1, 2, 3$. The minimizers

$$x(\mu) = \frac{\mu}{2}$$

were marked on each curve to illustrate how their positions change with the multiplier.

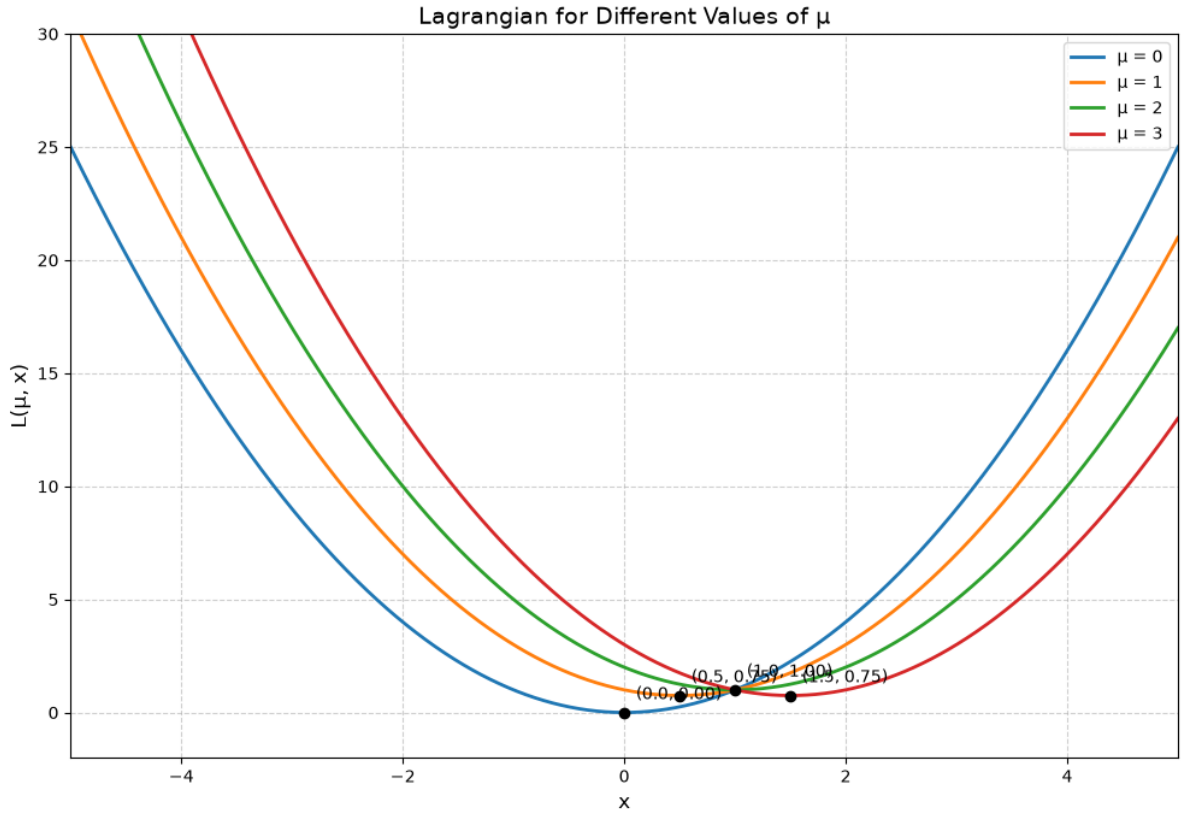


Figure 1: Lagrangian curves for $\mu = 0, 1, 2, 3$.

As μ increases, the minimizer moves to the right at a constant rate of $\frac{1}{2}$ unit per unit increase in μ. In particular, when $\mu = 2$, the Lagrangian minimizer reaches $x = 1$, which is exactly the primal optimal solution.

2.2 Deriving and plotting the dual function

The dual function is defined as

$$d(\mu) = \min_x \left[x^2 + \mu(1 - x)\right].$$

The minimizer is obtained by differentiating the Lagrangian with respect to x and setting the derivative to zero:

$$\frac{\partial L}{\partial x} = 2x - \mu = 0 \quad \Rightarrow \quad x = \frac{\mu}{2}.$$

Substituting this minimizer into the Lagrangian gives

$$d(\mu) = \left(\frac{\mu}{2}\right)^2 + \mu\left(1 - \frac{\mu}{2}\right).$$

Simplifying,

$$d(\mu) = \frac{\mu^2}{4} + \mu - \frac{\mu^2}{2}.$$

Therefore, the dual function is

$$\boxed{d(\mu) = \mu - \frac{\mu^2}{4}, \quad \mu \geq 0.}$$

It was then plotted against μ.

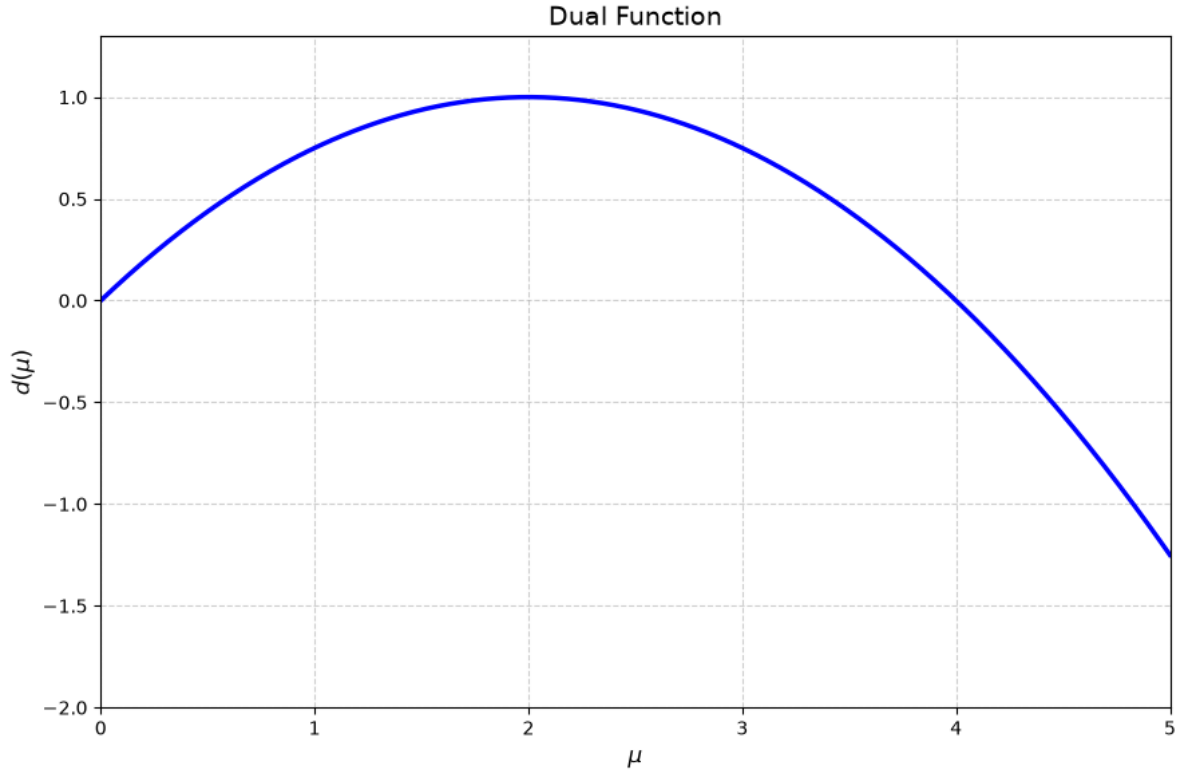


Figure 2: Dual function

The dual problem is to maximize the dual function over $\mu \geq 0$:

$$\max_{\mu \geq 0} d(\mu).$$

Given

$$d(\mu) = \mu - \frac{\mu^2}{4},$$

differentiating with respect to μ gives

$$\frac{dd(\mu)}{d\mu} = 1 - \frac{\mu}{2}.$$

Setting the derivative to zero,

$$1 - \frac{\mu}{2} = 0 \quad \Longrightarrow \quad \boxed{\mu^* = 2}.$$

$\mu^* = 2$ gives the maximum.

Substituting $\mu^* = 2$ into the dual function,

$$d(2) = 2 - \frac{2^2}{4} = 2 - 1 = 1.$$

Hence, the optimal dual solution and the maximum dual value are

$$\boxed{\mu^* = 2, \qquad d^* = 1}.$$

2.3 Deriving and plotting the primal function

The primal function

$$g(x) = \begin{cases} +\infty, & x < 1, \\ x^2, & x \geq 1 \end{cases}$$

was plotted over $x \in [-1, 3]$, showing the feasible and infeasible regions.

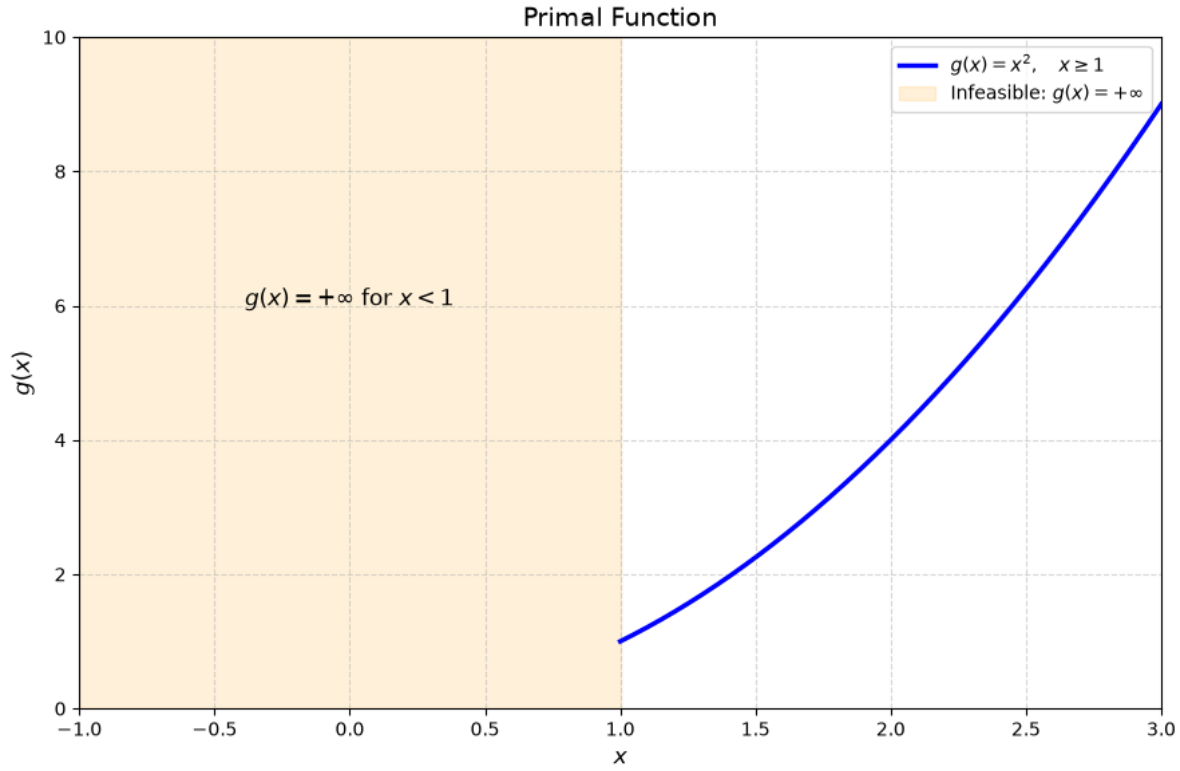


Figure 3: Primal function showing the feasible and infeasible regions

The primal function assigns an infinite value to every infeasible point ($x < 1$) because the Lagrangian becomes unbounded above as $\mu \to \infty$. For feasible points ($x \geq 1$), the primal function retains the original objective function $g(x) = x^2$. Consequently, minimizing $g(x)$ automatically excludes all infeasible points and is equivalent to the original constrained optimization problem. The minimum occurs at $x^* = 1$, giving the primal optimal value $p^* = g(1) = 1$.

3 Saddle Point and Complementary Slackness

3.1 Saddle-point condition

Computing the Lagrangian at the optimal values $x^* = 1$ and $\mu^* = 2$,

$$L(\mu^*, x^*) = L(2, 1) = 1^2 + 2(1 - 1) = 1.$$

Hence,

$$\boxed{L(\mu^*, x^*) = 1}.$$

Computing the Lagrangian at $x^* = 1$,

$$L(\mu, x^*) = L(\mu, 1) = 1^2 + \mu(1 - 1) = 1.$$

Since the constraint term $\mu(1 - 1) = 0$, the Lagrangian does not depend on μ. Therefore,

$$L(\mu, x^*) = L(\mu^*, x^*) = 1.$$

Hence, the left inequality holds with equality for all $\mu \geq 0$:

$$\boxed{L(\mu, x^*) \leq L(\mu^*, x^*)}.$$

Substituting $\mu^* = 2$ into the Lagrangian,

$$\begin{aligned} L(\mu^*, x) &= L(2, x) \\ &= x^2 + 2(1 - x) \\ &= x^2 - 2x + 2 \\ &= (x - 1)^2 + 1. \end{aligned}$$

Since $(x - 1)^2 \geq 0$, the minimum occurs at $x^* = 1$. Hence,

$$L(\mu^*, x) \geq 1 = L(\mu^*, x^*).$$

Therefore, the right inequality holds for every $x \in \mathbb{R}$:

$$\boxed{L(\mu^*, x^*) \leq L(\mu^*, x)}.$$

Combining the two inequalities,

$$\boxed{L(\mu, x^*) \leq L(\mu^*, x^*) \leq L(\mu^*, x)}$$

Hence, $(x^*, \mu^*) = (1, 2)$ is a saddle point of the Lagrangian.

The functions $L(\mu, x^*)$ and $L(\mu^*, x)$ were plotted against μ and x, respectively, with the common value $L(\mu^*, x^*) = 1$ marked on both graphs.

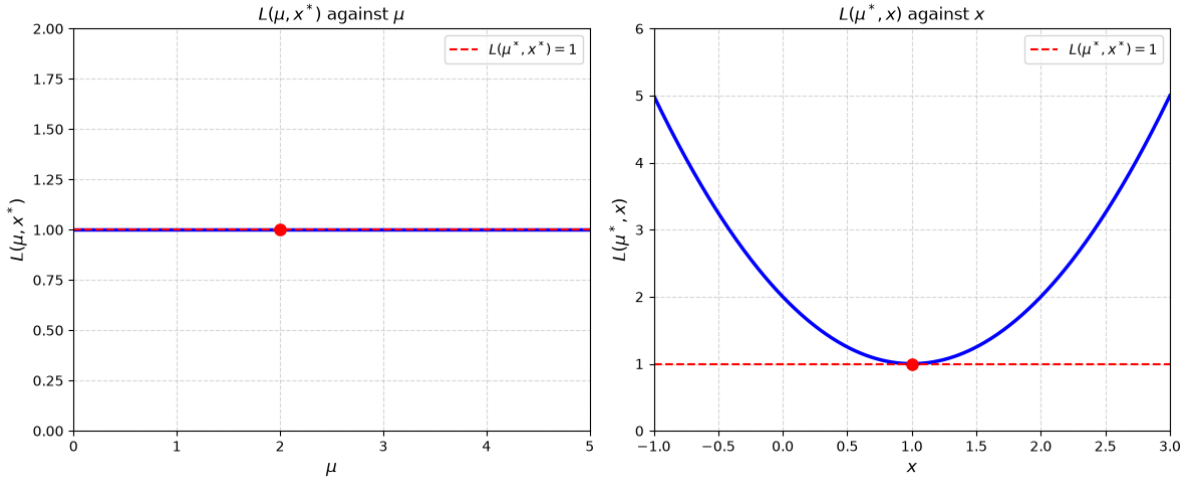


Figure 4: Plots of $L(\mu, x^*)$ and $L(\mu^*, x)$ illustrating the saddle-point condition

3.2 Explanation of the optimality argument

For any feasible x, we have

$$f(x^*) = L(\mu^*, x^*) \leq L(\mu^*, x) = f(x) + \mu^* h(x) \leq f(x).$$

Reasons for each step:

1. Since $h(x^*) = 0$, the constraint term $\mu^* h(x^*)$ becomes zero. Therefore, $f(x^*) = L(\mu^*, x^*)$.
2. Since $x^* = 1$ minimizes $L(\mu^*, x)$ for $\mu^* = 2$, its value cannot exceed the value at any other x.
3. By the definition of the Lagrangian, $L(\mu^*, x) = f(x) + \mu^* h(x)$.
4. Since $\mu^* \geq 0$ and $h(x) \leq 0$ for feasible x, we have $\mu^* h(x) \leq 0$. Hence, $f(x) + \mu^* h(x) \leq f(x)$.

The left inequality is

$$L(\mu, x^*) \leq L(\mu^*, x^*), \quad \forall \mu \geq 0.$$

Substituting the Lagrangian and cancelling $f(x^*)$ gives

$$\mu h(x^*) \leq \mu^* h(x^*).$$

If $h(x^*) > 0$, the left-hand side would increase without bound as $\mu \to \infty$, while the right-hand side remains constant. This would violate the inequality. Therefore,

$$\boxed{h(x^*) \leq 0}$$

3.3 Complementary slackness and the modified constraint

The complementary slackness condition is

$$\mu^* h(x^*) = 0.$$

For the original constraint, $h(x) = 1 - x$. At $x^* = 1$, the constraint function equals zero. Therefore,

$$\mu^* h(x^*) = 2(1 - 1) = 0.$$

Hence, the complementary slackness condition is satisfied. The constraint is active because equality holds at the optimum, and the corresponding multiplier is positive.

The new constrained optimization problem is

$$\min_x x^2 \quad \text{subject to } x \geq -1.$$

The objective function $f(x) = x^2$ attains its minimum at $x = 0$, where $f(x^*) = 0$. Since $0 \geq -1$, the solution satisfies the constraint. Therefore,

$$x^* = 0, \qquad f(x^*) = 0.$$

Writing the constraint as $h(x) = -1 - x \leq 0$, we obtain

$$h(x^*) = -1 - 0 = -1 < 0.$$

Hence, the constraint is inactive at the optimum.

The complementary slackness condition is

$$\mu^* h(x^*) = 0.$$

Substituting $h(x^*) = -1$ gives

$$-\mu^* = 0.$$

Therefore,

$$\boxed{\mu^* = 0}.$$

The optimal multiplier is zero because the constraint is inactive and does not restrict the optimal solution.

4 Weak Duality

4.1 Proof and numerical verification

Let $\mu \geq 0$ and let x be any feasible point, so that $h(x) \leq 0$. We have

$$d(\mu) \overset{(1)}{\leq} L(\mu, x) = f(x) + \mu h(x) \overset{(2)}{\leq} f(x).$$

The reasons for the two inequalities are as follows:

1. By definition, $d(\mu) = \min_x L(\mu, x)$. Hence, the dual function cannot exceed the value of the Lagrangian at any particular x.
2. Since $\mu \geq 0$ and $h(x) \leq 0$, we have $\mu h(x) \leq 0$. Therefore, $f(x) + \mu h(x) \leq f(x)$.

Combining these inequalities gives

$$d(\mu) \leq f(x)$$

for every feasible x. Since p^* is the minimum objective value over all feasible points, it follows that

$$\boxed{d(\mu) \leq p^*, \qquad \forall \mu \geq 0.}$$

This result is known as weak duality, which states that the dual function provides a lower bound on the optimal value of the primal problem.

For Question 1(b), the dual function and primal optimal value are

$$d(\mu) = \mu - \frac{\mu^2}{4}, \qquad p^* = 1.$$

Evaluating the dual function for the given values of μ gives

μ	$d(\mu)$	Verification
0	0	$0 \leq 1$
1	0.75	$0.75 \leq 1$
2	1	$1 \leq 1$
3	0.75	$0.75 \leq 1$

Thus, $d(\mu) \leq p^* = 1$ for all four values, verifying weak duality numerically.

4.2 Lower envelope of the Lagrangian lines

For a fixed value of x, the Lagrangian is a straight line as a function of μ. Substituting $x = 0, 1, 2$ gives

$$L(\mu, 0) = \mu, \qquad L(\mu, 1) = 1, \qquad L(\mu, 2) = 4 - \mu.$$

These functions were plotted against μ for $x = 0, 1, 2$, as shown below.

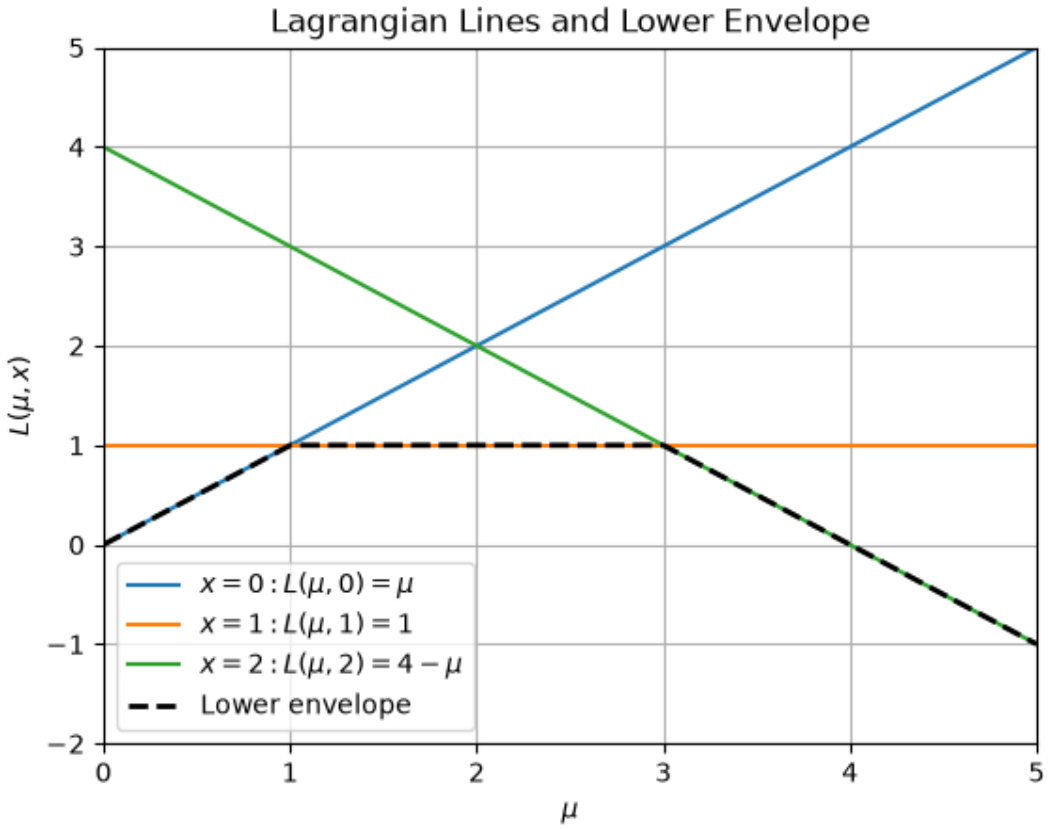


Figure 5: Lagrangian lines and their lower envelope

The lower envelope is concave because its slope decreases from $+1$ to 0 and then to -1. In general, the Lagrangian is affine in μ for each fixed x, and the pointwise minimum of affine functions is concave. Therefore, the dual function is concave even when $f(x)$ is non-convex.

5 Conclusion

The Lagrangian and dual functions were derived, and their optimal solutions were identified. The primal function demonstrated how infeasible points can be excluded from the optimization problem. The saddle-point condition, complementary slackness, and weak duality were verified. Finally, the primal and dual optimal values were both found to be 1, confirming that the problem has zero duality gap.